

Examiner reports

Q1.

Part (a) was often done well, but a number of candidates effectively repeated the same assumption twice. The candidates were often helped by the printed answer in part (b). Many of the candidates made life difficult for themselves by finding the initial speed and angle of projection and then working with these values. One consequence of this was that the final answers were inaccurate. Part (c) proved more challenging for candidates. The most common incorrect approach was to assume that the ball hit the wall when it was at its maximum height. In part (d) there were some good answers and many of the candidates were on the right track. However quite a few of the candidates produced answers which lacked clarity of explanation.

Q2.

Part (a) was frequently done well, with the most common error in part (a)(ii) being the introduction into the equation of motion of an additional force, usually the weight of the motorcycle and rider. Part (b) seldom produced assumptions backed by sound reasons. The majority of responses referred to information given in the question. In part (c), the most convincing solutions were based on the algebraic link between the force and the radius. There was a tendency in part (c) to assume that the magnitude of the frictional force could be altered by changing the radius of the circle, rather than the necessity to adjust the radius to compensate for the force.

Q3.

Many candidates proved competent in handling vector expressions, but a significant number lacked appreciation of vector and scalar quantities and the necessary notation. Errors in differentiation of trigonometrical functions, particularly in signs, also spoiled otherwise excellent solutions. Otherwise part (a) was done well, with the most frequent error being in the failure to present an answer in radians in part (a)(iii). Responses in part (b)(i) were variable, with scalar answers frequently presented, while in part (b)(ii), few appreciated the variable nature of the direction of the force, and many seeming unaware of both upper and lower bounds of the magnitude of the force.

Q4.

For those who were well prepared in the calculus techniques linking the quantities involved, this question proved highly successful. Part (c)(ii) proved discriminating, as expected, with only the most able candidates linking the components of the velocity vector with the given vector in an appropriate way.

Q5.

Candidates produced good work in this question, and were able to score marks on the parts where they had the necessary knowledge and understanding. Part (a)(i) was done well, and part (a)(ii), although an unfamiliar request, produced many good solutions. Part (a)(iii) was done well in a variety of ways. Parts (b) and (c) proved successful for most candidates.

Q6.

It was refreshing to see a wide variety of approaches to this question, which differed in many ways from previous questions on this topic. Many candidates were able to find the coordinates of the ball in part (a) and at this stage an error in the direction of the y-value was condoned. Part (b) provided interesting responses, using either the answers to part (a) with subsequent elimination of the time, or, in some cases, the full equation of the trajectory. In part (c) many candidates used the result of part (b), while others successfully found the time involved as an intermediate step. Part (d) proved to be more discriminating and there was confusion, for some, as to the quantities involved in the motion in the two directions. Many candidates reached the correct result following sound work, but some lost a mark through failing to adhere to the degree of accuracy required by the rubric for the paper. Overall it is encouraging to see candidates who are unable to complete one part of a question continue to other parts where they can be more successful.

Q7.

Most could attempt differentiation correctly to find an expression for the velocity in part (a).

However poor algebra and lack of appreciation of the form of scalar quantities let down many in part (b). Correct answers in part (b) were often then reduced to ' $6 + 2t$ ' in part (c) and few completed the request correctly.

Q8.

It should be noted that vector notation is essential throughout such questions as this, whether in the form of the unit vectors $\mathbf{i}$ and $\mathbf{j}$, or in column vector form. Part (a) was usually started well, but too often the constant of integration was missing. Part (c) tested understanding and proved discriminating as expected.

Q9.

Parts (a) and (b) were done well by the better prepared candidates, but few spotted and used the pattern of decreasing distances in part (c).

Q10.

The technique of integration was correctly selected and performed accurately by most candidates. Many then completed part (a) correctly, but a minority omitted the constant of integration and failed to use the initial conditions to establish the correct position vector. Part (b) was often fully correct, although some candidates added the position vectors of C and D rather than subtracting them in their attempt to find the vector $\overrightarrow{CD}$.

Q11.

The quality of solutions to problems set on this area of the specification has improved steadily since the first paper in 2001. It was pleasing to see a variety of methods being used and many candidates scored well on this demanding question.

Part (a) was attempted successfully either using the equations of constant acceleration or by integration. The most common error was an incorrect sign in the acceleration of the vertical component.

Part (b) proved to be the most demanding part of the question. Again, varied methods were used, ranging from a full verification of the result to the quotation of the general

equation of the path of the trajectory with appropriate substitutions.

Part (c) was popular, yielding marks to most candidates.

The varied methods used in part (d) revealed clear understanding of the motion. The most popular methods used the result quoted in part (b) or the equations of motion with constant acceleration, sometimes incorporating the symmetry of the motion.

In part (e), for those candidates who realised that the greatest speed occurred either at O or at C , the request was straightforward and full marks were often obtained.

Q12.

Part (a) showed an improvement in handling vectors in this form and also in differentiation techniques. Part (b) proved discriminating, as intended, and many failed to correctly link the direction of motion with the velocity components. Many candidates were successful in finding a correct expression for the acceleration in part (c), but not all were then able to correctly interpret their expression.

Q13.

Again the poor use of algebra proved to be the downfall of many candidates. Part (a) was answered quite well, with a satisfactory proportion of candidates reaching the printed solution, but there were many careless errors in part (b). Part (c) was generally successful, but few were able to apply the same principle in part (d), with the given speed often being treated as a vertical velocity component.

Q14.

Part (a) of this question was generally done very well, although some unusual vector manipulation was observed on a few scripts. Several candidates worked with each component and combined their results into a vector, rather than working with vectors throughout their solutions.

In part (b), a large number of candidates found the velocity correctly, but did not then go on to find the speed of the particle.

There were a fair number of good attempts at part (c). Some candidates made errors with the signs when first setting up their equations. There were a few candidates who did not make any significant attempt at this part of the question.

Q15.

Parts (a), (b) and (c) were generally answered very well.

In part (a), a few candidates tried to substitute $t = 0$ into the position vector without differentiating.

In part (c), some candidates found the acceleration, but did not go on to find the magnitude.

Part (d) was more challenging. Quite a few candidates stated that the velocity would become zero. There were some very good answers, but also quite a few that were somewhat confused.

Q16.

Part (a) was generally answered well with a mixture of integration and constant acceleration equation methods.

Part (b) was done better than expected, with only a small number of candidates working with the wrong component.

There were a variety of minor errors in part (c), but also many correct solutions.

Many candidates found part (d) too demanding and made little progress. It was not uncommon to see solutions that still contained an acceleration or that ignored the initial position of the particle.

Q17.

Many candidates were able to complete part (a), probably helped by the printed answer. Many of these candidates did not produce any further correct working. In part (b), some candidates could write down correct expressions for the position vector. A common error was to apply the constant acceleration formula $\mathbf{r} = \frac{1}{2}(\mathbf{u} + \mathbf{v})t$ using the two velocities given in the question without realising that $\mathbf{v}$ was variable.

Of the candidates who had a correct answer in part (b), a fair proportion went on to try part (c). There were a reasonable number of correct responses, but there were a number who used the wrong component or a mixture of the two. There were very few correct responses to part (d). The main incorrect approach was to use the position vector. Those candidates who realised they needed to find an expression for the velocity generally went on to complete this part of the question.

Q18.

Parts (a) and (b) proved to be quite accessible for most candidates, but part (c) was very much more demanding. In part (a) the candidates tended to score all or no marks, with a complete solution following a correct first step. There were occasional arithmetic errors in the working of some candidates. A correct answer to part (a) was often followed by the velocity in part (b). A few candidates used incorrect formulae or tried to use an expression for the position vector rather than the velocity. Part (c) caused a variety of problems, with some candidates forming equations that contained vectors. A few produced an equation that would have arisen from moving south west instead of south east. A correct equation was normally followed by a complete solution. A very small number of candidates found the velocity, but not the speed of the boat.

Q19.

The candidates who chose the appropriate constant acceleration equation in part (a) had little difficulty. However, a number did treat the position vector as if it were a velocity vector. In part (b), some candidates stopped when they had found the velocity and did not go on to find the speed as requested. A few candidates had problems dealing with the negative signs. Part (c) was very challenging. Some candidates made good attempts at this, but they were very much in the minority. There were quite a few reasonable answers to part (d).

Q20.

This question was done well by quite a few candidates. Many of these found the acceleration as part of their solution to part (a). The candidates who had difficulties were those who did not realise that they needed to use a constant acceleration formula or those who quoted an inappropriate formula as their starting point. A few candidates did have difficulty with part (b), but several who had struggled with part (a) used the printed result to enable them to attempt part (b). Those who had been successful with part (b) normally went on to complete part (c).

Q21.

Part (a) was done very well by a good number of the candidates. A few worked with the magnitudes of the quantities, starting with the magnitude of the force, rather than with the actual vectors. Part (b) was found to be much more demanding. The two main errors made by the candidates were to assume that the particle still had the same acceleration as in part (a) or to use a time of 45 s rather than 25 s for the second stage of the motion.

Q22.

Part (a) was answered correctly by almost all of the candidates. While some went on to deal correctly with part (b), a large number of candidates worked with the wrong component. Part (c) was found to be more challenging and there were very few correct solutions. Most of the candidates worked with the position vector instead of the velocity vector.

Q23.

Parts (a) and (b) were quite well answered; particularly part (a) where the answer was given.

In part (b) a few candidates confused the initial and final velocities. Some candidates found the position by assuming that at time t the velocity of the aeroplane would be $19\mathbf{i}+13\mathbf{j}$.

Part (c) caused a great deal of difficulty for some candidates. Many did not separate the two components of the position vector and tried to find t by working with the whole vector. One consequence of this was that final answers for the time were given as vectors. Those candidates who formed separate equations for each component of the position vector were more successful, but some did make errors solving the equations.

Q24.

A very small number of candidates seemed unsure how to deal with the vectors and made very little progress with the question. The majority of candidates did produce quite good solutions. Part (a) did not cause any particular difficulties. In part (b), some

candidates wrote $\sqrt{9^2 - 4^2}$ instead of $\sqrt{9^2 + 4^2}$. Those who could visualise part (c) had no difficulty, but some found it hard to identify a triangle to work in and tried to use the sine or cosine rules. A few candidates produced incorrect expressions involving sin or cos instead of using tan.

Q25.

Candidates found part (a) very straightforward, but had more difficulty with parts (b) and (c). Here, some did not know how to proceed and so did not make any real attempt at

these parts of the question. This was particularly true of part (c). In part (b), the most common incorrect approach was to use the equation $\mathbf{r} = \frac{1}{2}(\mathbf{u} + \mathbf{v})t$, but taking $\mathbf{v}$ to be the velocity at $t = 2$ and not $t = 3$. A significant number of candidates left their answer as a vector and did not find its magnitude. There were very few correct solutions to part (c), with many candidates not making any attempt at this part of the question. Of those that did, very few realised that they needed to work with the magnitude of the velocity rather than the velocity vector. Several candidates produced equations where 10 was put equal to a vector.

Q26.

Part (a) of this question was done well by almost all candidates. There were a lot of good solutions for the rest of the question, but there were two problems that emerged. The first was concerning the constants of integration. Some candidates did not include these in their working and simply added a 6 to the velocity component in part (b) to get the right answer. Candidates do need to show that they introduce constants when integrating and that they do evaluate them. The other problem emerged less frequently, but some candidates did try to use constant acceleration equations to find the velocity and position vectors.

Q27.

Part (a) was done very well, but part (b) proved to be more demanding with only a few candidates gaining many of the marks available on this part of the question. Most candidates failed to set up the quadratic equations needed to solve the problem. A few candidates tried to apply the quadratic equation formula to find t , but taking the coefficients as vectors, that is $a = -\mathbf{i} + 2\mathbf{j}$, $b = 4\mathbf{i} + 7\mathbf{j}$ and $c = -3\mathbf{i} - 5\mathbf{j}$. The candidates who realised how to approach the problem often gained full marks, although a few did run into difficulties having first stated the correct quadratic equations.

Q28.

The candidates found this question very demanding and there were very few complete solutions. Many candidates obtained marks of 0 or 1 on this question. While a good number of candidates could find the vector $\vec{AB}$, many candidates added the two vectors rather than subtracting. In the other parts of the question many candidates simply did not recognise how to proceed. Several tried to find the magnitudes of the vectors or use equations of the form as $v^2 = u^2 + 2as$. Some candidates made reasonable attempts at parts (c) and (d), but using an incorrect velocity from part (b). The candidates who produced good solutions were clearly very confident when working with vectors.

Q29.

Part (a) of this question was done well by many candidates. The main reason that candidates lost marks was because they did not differentiate correctly. For parts (b) and (c), many candidates used the position vector rather than the velocity vector as the basis for their working. This fundamental error was very common. In part (c), a number of candidates worked in degrees rather than radians, obtaining a final answer of 3600 instead of 20π .

Q30.

Many of the candidates found this question more difficult than expected. Part (a) was generally done well, but problems then emerged in part (b). Here many candidates found the magnitude of the acceleration and then used this scalar quantity in part (c). It was not uncommon to see an equation that involved a velocity that was a vector and an acceleration that was a scalar. Some of the candidates who worked through correctly stopped in part (c) and did not find the distance that had been requested.

Q31.

This question was quite challenging for the candidates and there were very few good solutions, although most candidates did pick up marks from one or two parts of the question. Part (a) was done by a fair number of candidates. Errors included treating the position at $t = 10$ as a velocity and not using this position vector at all. There were some good attempts at part (b) with candidates using their acceleration from part (a). Part (c) caused problems for many candidates. It was not uncommon to see answers where the time was given as a vector. Some candidates only considered one of the components. There was clearly a number of candidates who could write down a vector equation, but could not solve it to find T . In part (c)(ii), several candidates only found the velocity and did not go on to find the speed.



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